

Scenario Effects on Maximum Feasible Recycled Content

ΔPolicy = Rollback – Baseline | $\Delta\text{Capacity}$ = Decrease – Increase | $\Delta\text{Chemistry}$ = High-LFP – Benchmark | Cell: Δ change (Reference + Δ)

California
CA-origin feedstock / US recycling network

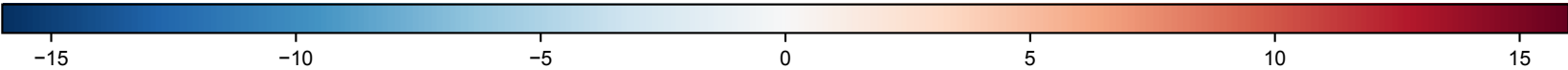
Ni	ΔPolicy	+1.8 (3.8)	+3.3 (9.0)	+5.6 (25.0)	-3.3 (30.0)	-6.7 (47.1)
	$\Delta\text{Capacity}$	+0.7 (2.6)	+3.7 (9.4)	+12.8 (32.2)	+11.8 (45.2)	+6.8 (60.6)
	$\Delta\text{Chemistry}$	+0.4 (2.3)	+1.9 (7.6)	+8.5 (27.9)	+9.2 (42.5)	+7.3 (61.1)
Mn	ΔPolicy	+1.8 (3.7)	+2.4 (6.8)	+3.8 (19.9)	-3.9 (29.1)	-6.8 (47.0)
	$\Delta\text{Capacity}$	+0.7 (2.5)	+2.8 (7.2)	+10.1 (26.3)	+11.4 (44.4)	+6.8 (60.6)
	$\Delta\text{Chemistry}$	+0.4 (2.2)	+1.5 (5.9)	+6.9 (23.0)	+8.9 (41.9)	+7.2 (61.0)
Co	ΔPolicy	+2.4 (4.9)	+4.0 (10.7)	+6.6 (29.9)	-4.0 (34.7)	-7.5 (51.1)
	$\Delta\text{Capacity}$	+0.9 (3.4)	+4.5 (11.2)	+15.5 (38.8)	+14.2 (52.9)	+7.8 (66.4)
	$\Delta\text{Chemistry}$	+0.5 (3.0)	+2.3 (9.1)	+10.4 (33.6)	+11.0 (49.7)	+8.2 (66.8)
Li	ΔPolicy	+1.3 (2.8)	+2.4 (7.0)	+4.5 (20.2)	-2.0 (26.4)	-4.6 (40.3)
	$\Delta\text{Capacity}$	+0.5 (2.1)	+2.7 (7.3)	+9.3 (24.9)	+9.2 (37.6)	+5.4 (50.3)
	$\Delta\text{Chemistry}$	+0.0 (1.6)	+0.1 (4.7)	+0.4 (16.0)	+0.4 (28.8)	+0.2 (45.2)
Gr	ΔPolicy	+1.3 (2.8)	+2.6 (7.5)	+4.9 (21.9)	-2.1 (29.3)	-5.0 (44.9)
	$\Delta\text{Capacity}$	+0.5 (2.1)	+2.9 (7.9)	+9.9 (27.0)	+10.0 (41.4)	+5.9 (55.8)
	$\Delta\text{Chemistry}$	-0.0 (1.6)	-0.1 (4.9)	-0.2 (16.8)	-0.2 (31.2)	-0.2 (49.8)
Cu	ΔPolicy	+1.3 (3.1)	+2.6 (8.0)	+5.1 (23.1)	-2.0 (31.5)	-5.0 (47.8)
	$\Delta\text{Capacity}$	+0.6 (2.3)	+2.9 (8.3)	+10.0 (28.0)	+10.3 (43.7)	+6.1 (58.8)
	$\Delta\text{Chemistry}$	-0.1 (1.7)	-0.4 (5.0)	-1.3 (16.6)	-1.6 (31.8)	-1.2 (51.6)
		2030	2035	2040	2045	2050

United States
US-origin feedstock / US recycling network

Policy	+1.2 (4.8)	+0.4 (6.0)	+0.1 (15.0)	-2.5 (26.9)	-3.2 (46.4)
	+1.3 (5.0)	+2.3 (7.9)	+7.8 (22.7)	+8.9 (38.3)	+5.7 (55.2)
	+0.7 (4.3)	+1.9 (7.5)	+6.3 (21.3)	+8.2 (37.5)	+7.3 (56.9)
Capacity	+0.3 (1.4)	+0.4 (3.2)	+0.1 (11.2)	-2.3 (24.8)	-3.0 (44.3)
	+0.4 (1.4)	+1.7 (4.5)	+6.5 (17.6)	+8.7 (35.8)	+5.6 (52.9)
	+0.2 (1.2)	+0.9 (3.7)	+4.4 (15.5)	+6.9 (34.0)	+6.1 (53.3)
Chemistry	+1.3 (5.2)	+0.5 (6.8)	+0.0 (17.8)	-3.0 (30.9)	-3.6 (50.4)
	+1.5 (5.4)	+2.7 (9.0)	+9.5 (27.2)	+10.7 (44.6)	+6.4 (60.4)
	+0.8 (4.7)	+2.2 (8.5)	+7.7 (25.4)	+9.7 (43.6)	+8.2 (62.2)
Policy	+1.2 (5.3)	+0.4 (6.4)	+0.6 (14.5)	-1.4 (25.3)	-2.0 (40.8)
	+1.5 (5.6)	+1.9 (7.9)	+5.7 (19.7)	+6.8 (33.6)	+4.4 (47.3)
	+0.1 (4.2)	+0.1 (6.1)	+0.3 (14.3)	+0.4 (27.1)	+0.3 (43.2)
Capacity	+0.2 (1.4)	+0.5 (4.3)	+0.8 (13.2)	-1.3 (25.2)	-2.0 (42.5)
	+0.3 (1.5)	+1.9 (5.7)	+6.1 (18.4)	+7.4 (34.0)	+4.9 (49.4)
	-0.0 (1.1)	-0.0 (3.7)	-0.1 (12.2)	-0.2 (26.4)	-0.2 (44.4)
Chemistry	+1.5 (7.2)	+0.5 (8.3)	+0.9 (18.1)	-1.3 (31.1)	-2.1 (49.2)
	+2.0 (7.6)	+2.2 (10.0)	+6.2 (23.4)	+7.6 (40.1)	+5.0 (56.3)
	-0.2 (5.4)	-0.5 (7.4)	-1.2 (16.0)	-1.6 (30.9)	-1.3 (49.9)
	2030	2035	2040	2045	2050

North America
NA-origin feedstock / NA recycling network

Policy	+1.4 (5.1)	+0.6 (6.4)	+0.3 (16.1)	-2.3 (29.4)	-3.4 (51.3)
	+1.4 (5.0)	+2.3 (8.1)	+8.1 (23.9)	+9.5 (41.3)	+6.2 (61.0)
	+0.7 (4.4)	+1.9 (7.7)	+6.6 (22.4)	+8.6 (40.4)	+7.8 (62.5)
Capacity	+0.4 (1.4)	+0.5 (3.4)	+0.3 (12.2)	-2.2 (27.2)	-3.3 (49.1)
	+0.3 (1.4)	+1.7 (4.7)	+6.8 (18.7)	+9.4 (38.8)	+6.2 (58.5)
	+0.2 (1.2)	+0.9 (3.8)	+4.6 (16.6)	+7.3 (36.8)	+6.5 (58.9)
Chemistry	+1.6 (5.5)	+0.7 (7.2)	+0.4 (19.2)	-2.8 (33.9)	-3.8 (55.8)
	+1.5 (5.4)	+2.8 (9.3)	+9.9 (28.7)	+11.4 (48.1)	+7.1 (66.7)
	+0.8 (4.7)	+2.2 (8.7)	+8.0 (26.8)	+10.3 (46.9)	+8.8 (68.5)
Policy	+1.5 (5.7)	+0.6 (6.7)	+0.9 (15.4)	-1.2 (27.2)	-1.9 (44.6)
	+1.5 (5.7)	+1.9 (8.0)	+5.9 (20.5)	+7.2 (35.6)	+4.7 (51.2)
	+0.1 (4.2)	+0.1 (6.2)	+0.3 (14.9)	+0.4 (28.8)	+0.3 (46.8)
Capacity	+0.3 (1.4)	+0.7 (4.4)	+1.0 (13.9)	-1.2 (27.1)	-1.9 (46.4)
	+0.3 (1.4)	+1.9 (5.7)	+6.3 (19.2)	+7.8 (36.1)	+5.2 (53.5)
	-0.0 (1.1)	-0.0 (3.7)	-0.1 (12.7)	-0.2 (28.1)	-0.2 (48.2)
Chemistry	+2.1 (7.8)	+0.8 (8.8)	+1.3 (19.1)	-1.0 (33.3)	-1.7 (53.5)
	+2.0 (7.7)	+2.2 (10.2)	+6.5 (24.3)	+8.0 (42.3)	+5.3 (60.5)
	-0.2 (5.5)	-0.5 (7.5)	-1.2 (16.6)	-1.6 (32.7)	-1.3 (53.9)
	2030	2035	2040	2045	2050



Δ maximum feasible recycled content (percentage points)